

1 ZebraPuzzle Example

Setup

 Houses	1, 2, 3
 Person	Alice, Bob, Carol
 Pet	Cat, Dog, Fish
 Drink	Tea, Coffee, Juice

Original Natural-Language Clues

- C1.** Alice lives immediately to the left of the cat owner.
- C2.** Bob does not drink coffee.
- C3.** House 2 serves tea.

2 Parsed Syntactic Clues

Clues converted to solver-checkable form

C1: Alice + 1 == Cat.

C2: Bob != Coffee.

C3: Tea == 2.

Alice, Bob, Carol → 1, 2, 3

Cat, Dog, Fish → Cat, Dog, Fish

Tea, Coffee, Juice → Tea, Coffee, Juice

Notation: + 1 means immediately to the right; == means "is in"; != means "is not in".

3 Parsed Reasoning Steps

R1 From C1, Alice cannot be in House 3.

S1 Alice != 3.

R2 From C3, House 2 serves tea.

S2 Tea == 2.

R3 Therefore House 2 cannot serve coffee.

S3 Coffee != 2.

R4 Bob drinks coffee.

S4 Bob == Coffee.

R5 Bob cannot be the tea drinker.

S5 Bob != 2.

 R: Natural language reasoning

 S: Syntactic (solver-checkable) step

4 Solver Evaluation

Step	Status	Why
S1	Valid, Novel	Follows from left-of relation.
S2	Valid, but Clue Restatement	Directly from C3.
S3	Valid, Novel	Derived from Tea == 2.
S4	Contradictory	Conflicts with C2 (Bob does not drink coffee).
S5	Valid, Novel	Combines C2 and C3.

 Valid, Novel (syntactic + informative)

 Valid, but Clue Restatement (non-novel)

 Contradictory

5 Process Reward Signal

 + reward for Valid + Novel steps

 0 or small reward for Valid but non-novel steps

 - penalty for Contradictory steps

 Valid steps: 4

 Novel steps: 3

 Contradictions: 1

Final process reward: **Positive**

Current solution (illustrative)

House	Person	Drink	Pet
1	Alice	–	–
2	–	Tea	–
3	Bob	Coffee	–

Reasoning trajectory is shaped using solver feedback.